

# Stylistic-STORM: Self-Supervised Spectral Disentanglement Using Adversarial Learning and JEPA for Weather Analysis

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P1.T1

MONDAY  
17 AUGUST 2026

15:00–16:30

Poster Session 1

CENTRAL IDEA

Appearance is not a nuisance to remove — it is semantic information to preserve.

## 1 WHY THIS MATTERS

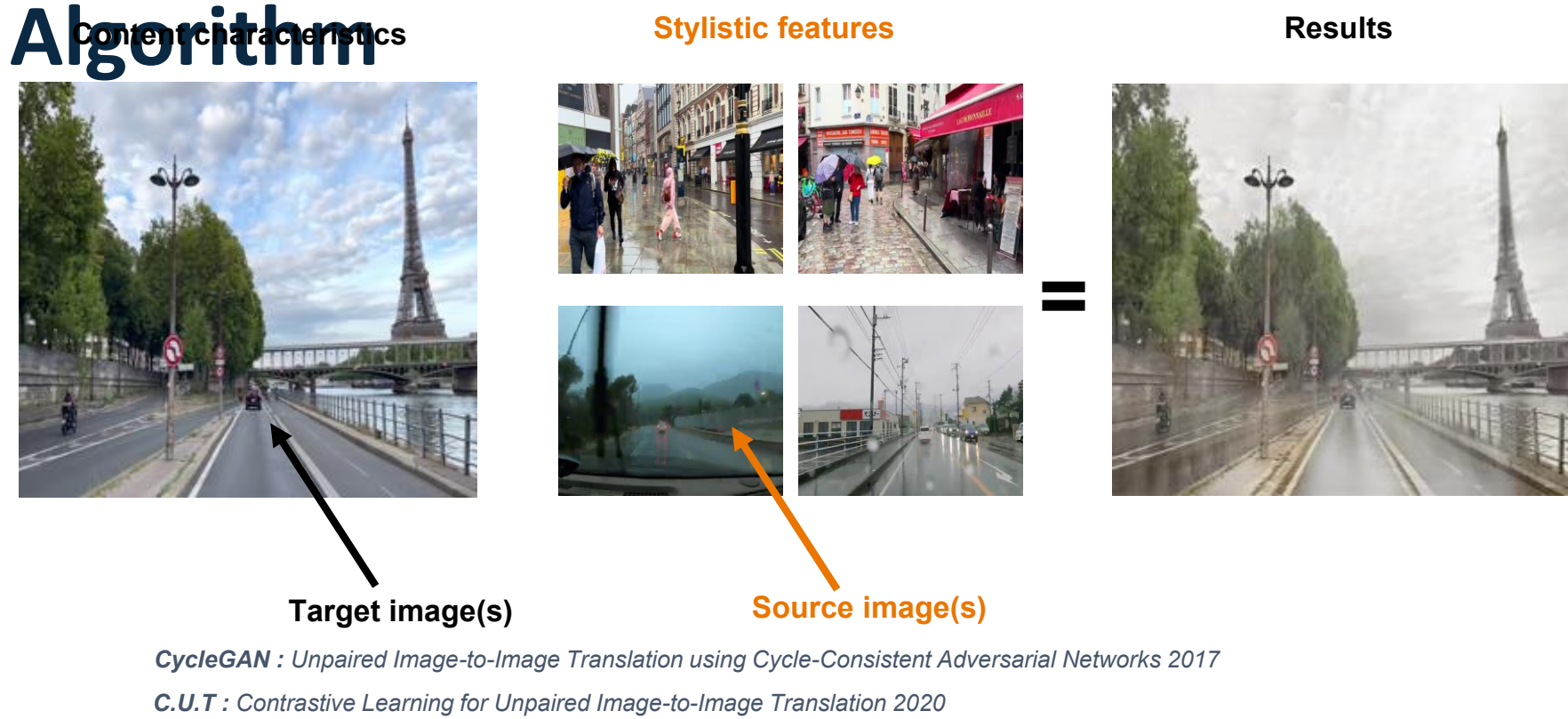
Most self-supervised methods learn invariance to appearance changes. That is useful for object identity—but risky when appearance carries the task signal.

In weather analysis, style is evidence

- rain streaks and droplets
- snow granularity and frequency signatures
- fog scattering, halos and contrast loss
- reflections and wet-road appearance

Can we preserve appearance without sacrificing semantic content?

### Style Transfer Algorithm



### Three contributions

#### 01 Two gated latent streams

Content remains stable; Style captures appearance signatures.

#### 02 Spectral + adversarial learning

Controlled perturbations expose textures, contrast and scattering.

#### 03 Predictable style tokens

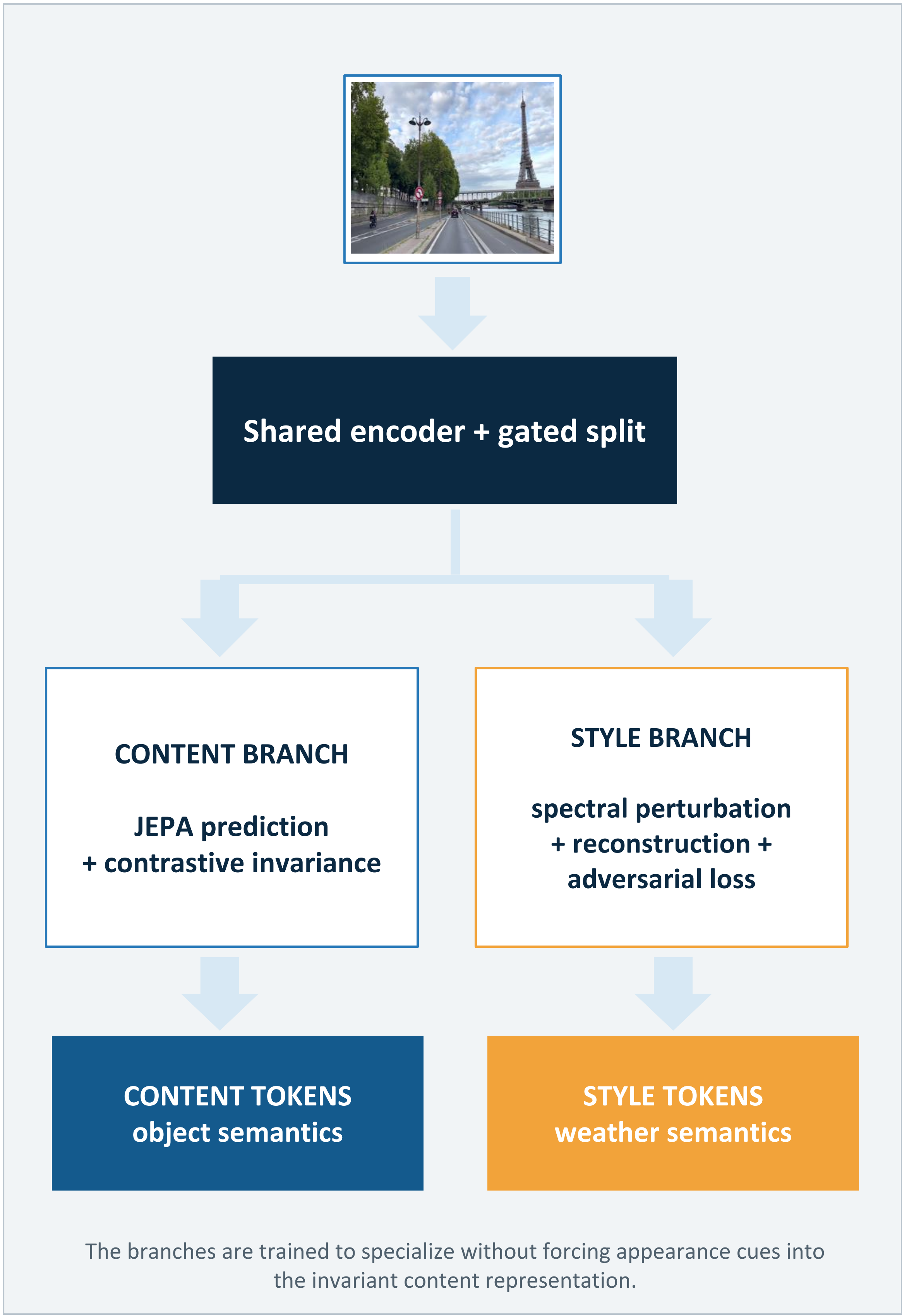
Style-JEPA filters contingent details into reusable representations.

#### Evaluation protocol

Self-supervised pre-training on 250k weather images  
↓  
Frozen representation + 10% labeled data  
↓  
Fine-grained weather characterization

## 2 ST-STORM ARCHITECTURE

A shared encoder is split by gating into complementary Content and Style learning objectives.



### Learning objectives

#### CONTENT

$L_{content} = L_{JEPA} + L_{contrastive}$

Stable semantics across appearance changes.

#### STYLE

$L_{style} = L_{pred} + L_{rec} + L_{adv}$

Predictable, reconstructible appearance signatures.

### Training cycle



## 3 EVIDENCE

The Style branch retains weather cues that invariant SSL baselines tend to suppress.

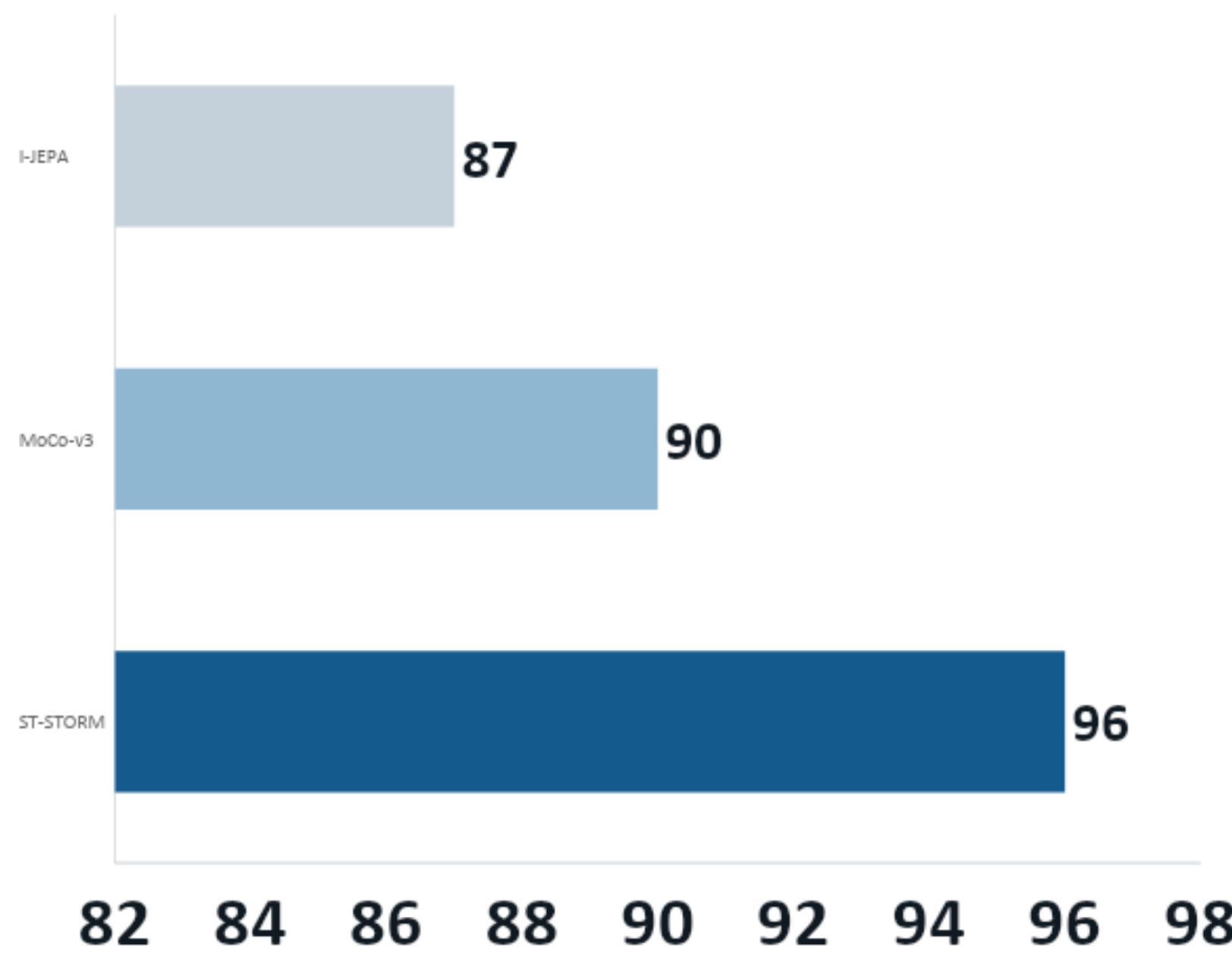
96%

global F1 on fine-grained weather characterization

10%

of labeled data used after self-supervised pre-training

### Frozen-feature comparison



### What the style tokens encode

#### TYPE

rain · fog · snow · clear

#### INTENSITY

severity and visibility

#### GROUND

dry · wet · snowy road

#### SPECTRUM

texture and frequency cues

#### TAKE-HOME MESSAGE

Disentangling style from content preserves critical appearance information while maintaining a robust semantic representation.

#### PAPER



#### CODE



Innovate UK



Co-funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or European Climate, Infrastructure and Environment Executive Agency (CINEA). Neither the European Union nor the granting authority can be held responsible for them. Project grant no. 101069576. UK participants in this project are co-funded by Innovate UK under contract no.10045139. Swiss participants in this project are co-funded by the Swiss State Secretariat for Education, Research and Innovation (SERI) under contract no. 22.00123.